

Quantifying emergence in systems of cities multiscale simulation models

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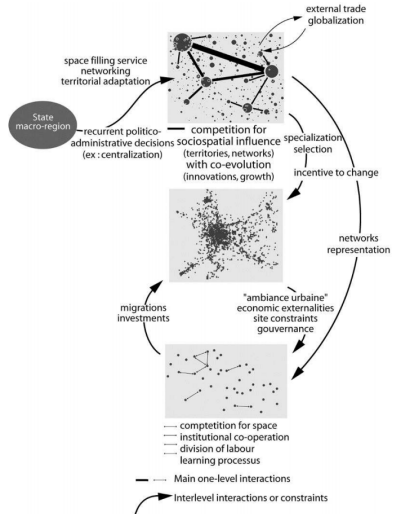
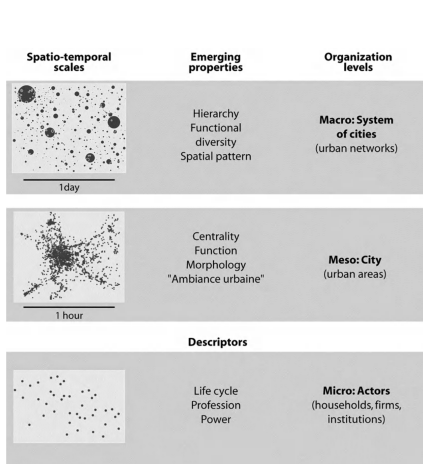
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Scales in Systems of Cities



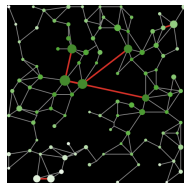
An evolutionary urban theory for cities as systems within systems of cities
[Pumain, 2018] [Pumain, 1997]

Systems of cities simulation models

- Agent-based models with economic processes: Simpop1 [Sanders et al., 1997], Simpop2 [Bretagnolle et al., 2006], Marius [Cottineau et al., 2015]
- Diffusion of innovations: SimpopLocal [Schmitt et al., 2015], innovation cycles [Favaro and Pumain, 2011], urban evolution [Raimbault, 2020c]
- Transport infrastructure: SimpopNet [Schmitt, 2014], Network effects [Raimbault, 2020b], Co-evolution [Raimbault, 2021a]
- Multi-modelling for sustainability trade-offs [Raimbault and Pumain, 2022]



Simpop1



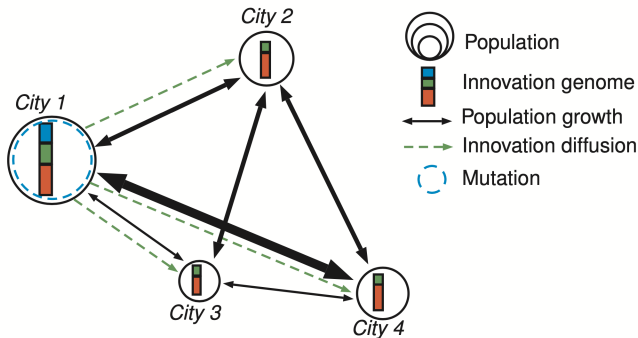
SimpopNet

Towards applied multiscalar models

- multi-scalar simulation models are needed to explore sustainable policies (territorial specificities, processes across scales, . . .) [Rozenblat and Pumain, 2018]
- stylised multi-scale urban models focusing on (i) urban form and interurban interactions [Raimbault, 2021c]; (ii) innovation diffusion within and between cities [Raimbault and Pumain, 2023]
- multi-scalar models must capture bottom-up and top-down feedbacks and a certain independence between ontologies at each scale [Raimbault, 2023], but how can they behave regarding the type of emergence? (downward causation, causal decoupling [Bedau, 2002])

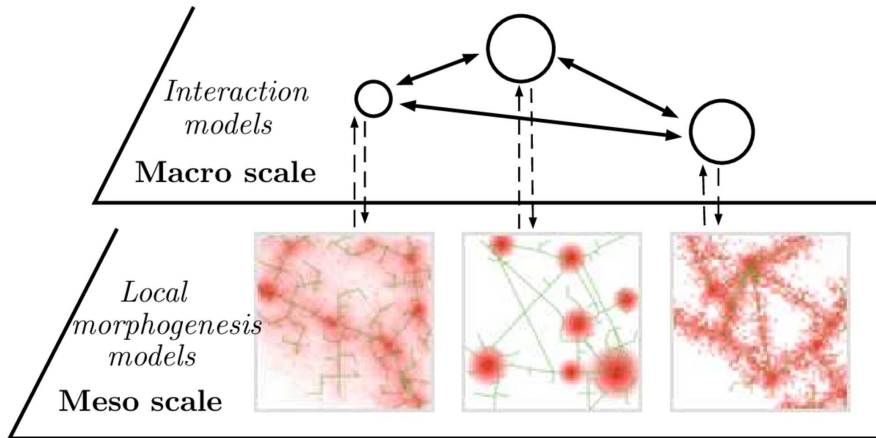
This contribution: *systematically quantify regimes of emergence with metrics from [Rosas et al., 2020], using a diversity search algorithm [Chérel et al., 2015] on two stylised multi-scalar urban simulation models [Raimbault, 2021c][Raimbault and Pumain, 2023].*

Model 1: multi-scale innovation dynamics



Model by [Raimbault and Pumain, 2023] coupling urban evolution based on innovation diffusion at the macro scale [Raimbault, 2020c], with intra-urban innovation cluster dynamics between firms for the emergence of new innovations [Raimbault, 2022]

Model 2: urban morphology and inter-city interactions



Strong coupling by [Raimbault, 2021c] between urban morphogenesis aggregation-diffusion processes at the meso scale [Raimbault, 2018] and interactions between cities at the macro scale [Raimbault, 2020b]

Indicators to quantify emergence

Quantification of emergence using information-theoretic indicators proposed by [Rosas et al., 2020]:

With V_t time-serie of a macro observable, X_t^j time-series of micro observables, I an estimator of mutual information, τ a delay:

$$\Psi_\tau(V) = I(V_t; V_{t+\tau}) - \sum_j I(X_t^j; V_{t+\tau}) \quad (1)$$

$$\Delta_\tau(V) = \max_j \left(I(V_t; X_{t+\tau}^j) - \sum_i I(X_t^i; X_{t+\tau}^j) \right) \quad (2)$$

$$\Gamma_\tau(V) = \max_j I(V_t; X_{t+\tau}^j) \quad (3)$$

$\Psi > 0$: causal emergence; $\Delta > 0$: downward causation; $\Gamma = 0$: causal decoupling

Implementation and experiments

→ models implemented in scala

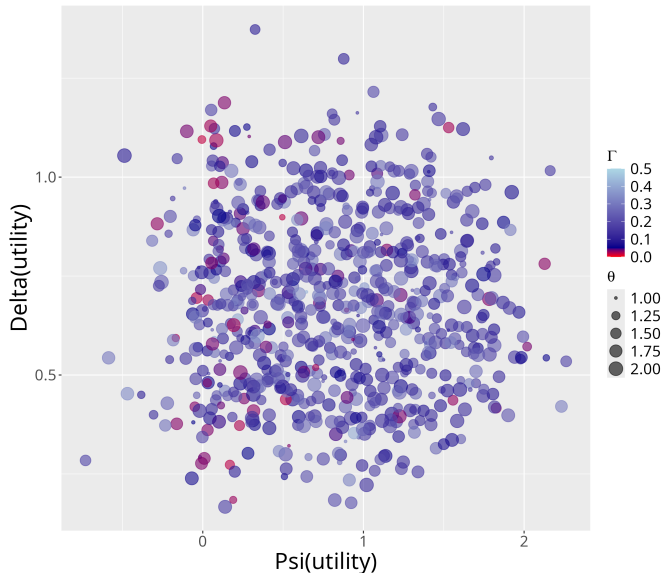
→ integration into the OpenMOLE model exploration and validation open source software [Reuillon et al., 2013] (*seamless (i) model embedding; (ii) access to HPC resources; (iii) exploration and optimisation algorithms*)



→ both model initialised on synthetic systems of cities with realistic features (Zipf's rank-size law, Clarke's law, spatial distribution) [Raimbault et al., 2019]

→ run the PSE algorithm [Chérel et al., 2015] on emergence indicators

Results: innovation regimes in the innovation model

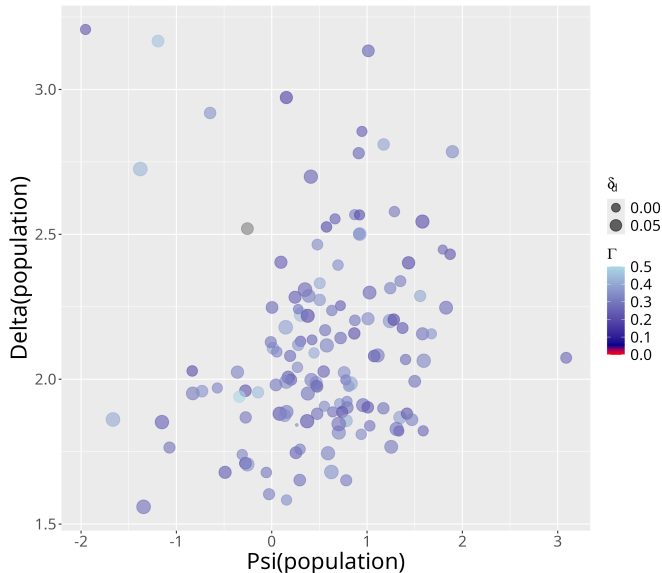


→ most regimes with $\Psi > 0$ and $\Delta > 0$
effective emergence
with downward
causation

→ difficult to get
 $\Delta \simeq 0$ (min 0.17)

→ causal decoupling
 $\Gamma \simeq 0$ occurs at
various Ψ, Δ values

Results: innovation regimes in the meso-macro model



→ similarly most regimes with emergence and downward causation

→ minimal Δ even higher (min 1.56): more structural constraints

→ no regime with causal decoupling

Discussion

- models capture effective emergence with downward causation, with a great diversity of regimes including causal decoupling
- both models behave differently, with stronger constraints for the meso-macro model (no causal decoupling)

Work in progress:

- extension of the method to a micro-meso simulation model with developpers and road network morphogenesis [Raimbault, 2021b] [Raimbault et al., 2014]
- statistical tests, more explorations and optimisations, link between parameter values and emergence regimes

Future work:

- test of the method on observed time-series and on data-parametrised models
- systematic review of inter-scale processes to build new models around these

Open repository:

<https://github.com/JusteRaimbault/EmergenceSystemsOfCities>



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Mesoscopic dynamics from [Raimbault, 2022]: **Innovation clusters** composed of firms, sets of employees (real-valued genome representing ideas); aim at optimising a fixed random fitness (Rastrigin function).

Meso cycle: in each innovation cluster (city) for t_m meso time steps:

- employee exchange ideas within firms (genome crossovers)
- idea with best fitness is chosen by each company
- ideas are informally exchanged between firms

Iterated steps, for t_f macro time steps:

- 1 Mesoscopic cycle
- 2 **Bottom-up feedback:** if relative fitness gain of the best firm exceeds a parameter ($\delta f_i > \theta$), the corresponding city will innovate at the macro scale.
- 3 Macroscopic step: innovation diffusion; population migration and growth; new innovations.
- 4 **Top-down feedback:** firm strategy is updated by changing crossover probability and mutation probability, and the urban environment in terms of informal exchanges, both depending on relative population growth.

- Macro observables: utility and diversity of innovations
- Meso observables: best fitness and diversity of ideas

Model setup on synthetic systems of cities resembling real systems
[Raimbault et al., 2019]

Model scales and objects:

- Cities as agents at the macroscopic scale
- Cities as population density grids at the mesoscopic scale

Processes:

- Interaction flows between cities and endogenous growth
- Urban sprawl and aggregation within urban areas
- **Downward feedback** Adaptation of urban form parameters to population and accessibility growth (“policy response”)
- **Upward feedback** Adaptation of city interaction range depending on internal performance (congested flows)

Meso-macro model formalization

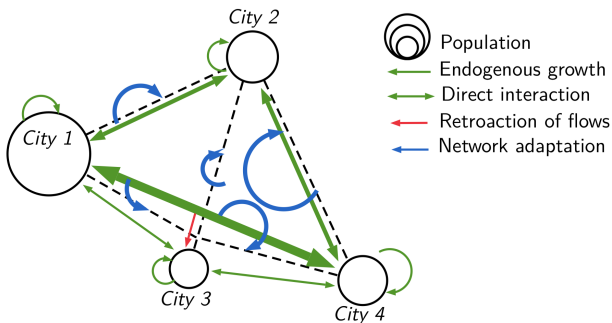
At each time step:

- ➊ Macroscopic population are evolved with the interaction model
- ➋ Population and accessibility differences modify the mesoscopic parameter of diffusion (sprawl) and aggregation (metropolization) (*downward feedback*)
- ➌ Urban forms are evolved for each city at the mesoscopic scale to match population increases
- ➍ City interaction ranges are modified according to the internal performance of each city (congested flows) (*upward feedback*)

Meso-macro: macroscopic model

System of cities interaction model including network evolution; production of multiple co-evolution regimes and calibration for France.

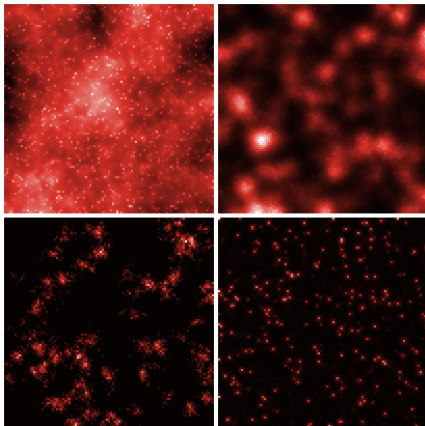
[Raimbault, 2020b]



Indicators at the macroscopic scale: distributions of population, accessibilities, centralities (summarized by average, hierarchy, entropy)

Meso-macro : mesoscopic model

Aggregation-diffusion model at the mesoscopic scale, covering most existing urban forms in Europe [Raimbault, 2018]



Indicators at the mesoscopic scale: urban form captured by Moran index, average distance, hierarchy, entropy

Meso-macro: model parameters

Type	Parameter	Process	Range
Macro	$g_i = g_0$	Endogenous growth	[0; 0.05]
	$w_i = w_G$	Interactions weight	[0; 0.01]
	$\gamma_i = \gamma_G$	Interactions hierarchy	[0; 5]
	d_i	Interactions decay	[0; 1000]
Meso	α_i	Aggregation	[0; 5]
	β_i	Diffusion	[0; 0.1]
	t_m	Urban growth speed	{1; 20}
	n_d	Diffusion steps	[1; 5]
Multiscale	$\delta\alpha$	Metropolization (downward)	[-0.2; 0.2]
	$\delta\beta$	Sprawl (downward)	[-0.2; 0.2]
	δd	Efficiency (upward)	[-0.2; 0.2]
	λ	Congestion cost (upward)	[0; 10]

Meso-macro: synthetic configuration

Model applied on synthetic systems of cities:

- random positions and rank-size hierarchy ($\alpha = 1.0$ and $P_0 = 100,000$)
- countrywide urban system scale: 500km and 20 cities
- initial population grids as monocentric (grid of size 50 and center cell density 1000 units)
- simulated for 20 macroscopic time steps (order of magnitude of half a century)